

The `tsd` Discrepancy Between `pmsimstats-orig` and `pmsimstats2025`

Summary

The `tsd` (time since discontinuation) variable in `carryover_power_simulation.R` (`pmsimstats2025`) is computed differently from the original `buildtrialdesign()` in `pmsimstats-orig`. The 2025 code produces `tsd = 0` at every first off-drug measurement, while the original code produces `tsd >= 1` (the interval width leading into that measurement). This discrepancy is the sole cause of the $t_{sd} = 0$ discontinuity documented in `predictor_comparison.tex`. The discontinuity does not exist in Hendrickson's original implementation.

The Two Algorithms

Original: `buildtrialdesign()` (`pmsimstats-orig`)

File: `01c-pmsimstats-orig/pmsimstats/R/buildtrialdesign.R`, lines 86–106.

The original computes `tsd` as the cumulative duration of off-drug intervals, measured in interval widths (`t_wk`):

```
t_wk <- c(4, 4, 1, 1, 1, 1, 4, 4)
tsd <- t_wk * (od != 1)
for (iTP in 2:nP) {
  if (od_intervals[iTP] > 0)
    tod[iTP] <- tod[iTP] + tod[iTP-1]
  else
    tsd[iTP] <- tsd[iTP] + tsd[iTP-1]
}
tsd <- everondrug * tsd
```

At the first off-drug timepoint, `tsd` equals the interval width leading into that timepoint (1 week for weeks 10–12 in the clustered phase, 4 weeks for the spaced phase). It is never zero.

The implicit assumption: the drug was discontinued at the *start* of the interval, so by the measurement at the end of the interval, the participant has been off drug for the full interval duration.

`pmsimstats2025: carryover_power_simulation.R`

File: `carryover_focus/carryover_power_simulation.R`, lines 332–343.

The 2025 code computes `tsd` as the difference between the current calendar week and the week at which discontinuation was detected:

```
just_discontinued = treatment == 0 &
  lag(treatment, default = 0) == 1
discontinuation_week = if_else(
  just_discontinued, week, NA_real_
)
discontinuation_week = zoo::na.locf(
  discontinuation_week, na.rm = FALSE
)
tsd = if_else(
  treatment == 0 & !is.na(discontinuation_week),
  pmax(week - discontinuation_week, local_tsd_min),
  0
)
```

At the first off-drug timepoint, `week == discontinuation_week`, so `tsd = 0`. The implicit assumption: discontinuation occurs *at* the measurement week, not before it.

The Hybrid Design

Both implementations use the same measurement weeks and path definitions. The Hybrid design (called ‘primary N-of-1 design’ in Hendrickson) has:

- **Measurement weeks:** 4, 8, 9, 10, 11, 12, 16, 20
- **Interval widths (t_{wk}):** 4, 4, 1, 1, 1, 1, 4, 4

The four ondrug vectors from the vignette (`Produce_Publication_Results_1_generate_data.Rmd`, lines 54–60):

Path A: `c(1, 1, 1, 1, 0, 0, 1, 0)`

Path B: `c(1, 1, 1, 1, 0, 0, 0, 1)`

Path C: `c(1, 1, 1, 0, 0, 0, 1, 0)`

Path D: `c(1, 1, 1, 0, 0, 0, 0, 1)`

These correspond to `pmsimstats2025` Paths 1–4 respectively, confirmed by matching the `case_when` logic in `create_hybrid_design()` at the three decision points (weeks 10, 16, 20).

Path-by-Path Comparison

In the tables below, ‘Orig’ refers to the original `buildtrialdesign()` output and ‘2025’ refers to `carry-over_power_simulation.R`. Only off-drug rows are shown (on-drug rows have `tsd = 0` in both implementations).

Path A / Path 1: ondrug = `c(1, 1, 1, 1, 0, 0, 1, 0)`

Week	ondrug	Orig tsd	2025 tsd	Match
4	1	0	0	Yes
8	1	0	0	Yes
9	1	0	0	Yes
10	1	0	0	Yes
11	0	1	0	No
12	0	2	1	No
16	1	0	0	Yes
20	0	4	0	No

Path B / Path 2: ondrug = `c(1, 1, 1, 1, 0, 0, 0, 1)`

Week	ondrug	Orig tsd	2025 tsd	Match
4	1	0	0	Yes
8	1	0	0	Yes
9	1	0	0	Yes
10	1	0	0	Yes
11	0	1	0	No
12	0	2	1	No
16	0	6	5	No
20	1	0	0	Yes

Path C / Path 3: ondrug = `c(1, 1, 1, 0, 0, 0, 1, 0)`

Week	ondrug	Orig tsd	2025 tsd	Match
4	1	0	0	Yes

Week	ondrug	Orig tsd	2025 tsd	Match
8	1	0	0	Yes
9	1	0	0	Yes
10	0	1	0	No
11	0	2	1	No
12	0	3	2	No
16	1	0	0	Yes
20	0	4	0	No

Path D / Path 4: **ondrug = c(1, 1, 1, 0, 0, 0, 0, 1)**

Week	ondrug	Orig tsd	2025 tsd	Match
4	1	0	0	Yes
8	1	0	0	Yes
9	1	0	0	Yes
10	0	1	0	No
11	0	2	1	No
12	0	3	2	No
16	0	7	6	No
20	1	0	0	Yes

The Pattern

The 2025 **tsd** is systematically shorter than the original by exactly one interval width at the first off-drug measurement after each discontinuation, and by the same offset at all subsequent off-drug measurements in that sequence.

The rule:

- **Original:** First off-drug **tsd** = interval width into that timepoint (1 week in the clustered phase, 4 weeks in the spaced phase). Subsequent off-drug measurements accumulate from there.
- **2025:** First off-drug **tsd** = 0 (because **week - discontinuation_week = 0**). Subsequent off-drug measurements are offset by the same initial deficit.

The offset is constant within each off-drug run but varies across discontinuation events depending on the interval width:

- Weeks 10–12 (clustered phase, **t_wk** = 1): offset = 1 week
- Week 16 (spaced phase, **t_wk** = 4): offset = 4 weeks
- Week 20 (spaced phase, **t_wk** = 4): offset = 4 weeks

Consequence for Dbc

The **Dbc** predictor is computed identically in both implementations:

$$\text{Dbc} = \left(\frac{1}{2}\right)^{\text{tsd}/t_{1/2}}$$

The different **tsd** values produce dramatically different **Dbc** at boundary measurements. For a Path C / Path 3 participant at $t_{1/2} = 0.1$ weeks:

Week	Orig tsd	Orig Dbc	2025 tsd	2025 Dbc
10	1	$(1/2)^{10} \approx 0.001$	0	1.000
11	2	$(1/2)^{20} \approx 0$	1	0.001
12	3	$(1/2)^{30} \approx 0$	2	≈ 0
20	4	$(1/2)^{40} \approx 0$	0	1.000

At the longer half-life $t_{1/2} = 0.2$ weeks:

Week	Orig tsd	Orig Dbc	2025 tsd	2025 Dbc
10	1	$(1/2)^5 = 0.031$	0	1.000
11	2	$(1/2)^{10} \approx 0.001$	1	0.031
12	3	$(1/2)^{15} \approx 0$	2	0.001
20	4	$(1/2)^{20} \approx 0$	0	1.000

The original produces near-zero carryover at boundary measurements for short half-lives. The 2025 code produces full carryover (**Dbc** = 1.000) at every first off-drug timepoint regardless of half-life.

Two Separate Issues

The **tsd** discrepancy and the published power behavior are related but distinct problems. They must be addressed in sequence.

Issue 1: The 2025 reimplementation does not match the original

The $t_{sd} = 0$ step-function power cliff documented in `predictor_comparison.tex` is caused by the 2025 **tsd** computation, not by Hendrickson’s original logic. Under the original computation:

- Boundary **tsd** is always at least 1 week (clustered phase) or 4 weeks (spaced phase).
- For $t_{1/2} = 0.1$ weeks, boundary **Dbc** ≈ 0.001 (not 1.000).
- The step-function artifact does not arise because the boundary measurements carry negligible carryover contamination.

Required fix: Adopt the original interval-based **tsd** computation so that `pmsimstats2025` faithfully reproduces the Hendrickson simulation as a baseline. This is a prerequisite for any further analysis of power behavior.

Issue 2: The original itself shows excessive power dropoff

Independent of the 2025 bug, the published Hendrickson heatmaps show power declining more rapidly with increasing carryover half-life than pharmacological reasoning would predict. This is the motivating observation for the entire `carryover_focus` analysis.

Once the 2025 **tsd** is corrected to match the original, the excessive power dropoff (if it persists) must have a different mechanism than the $t_{sd} = 0$ discontinuity. Candidate explanations include:

1. **Scale factor mismatch in the original code.** `generateData()` applies `scalefactor = 2` to the carryover decay in the DGP (line 96: $(1/2)^{(\text{scalefactor} * \text{tsd} / \text{t_half})}$), while `lme_analysis()` uses `carryover_scalefactor = 1` when computing **Dbc** (line 109: $(1/2)^{(\text{op}\$carryover_scalefactor * \text{tsd} / \text{op}\$carryover_t1half)}$). This means the DGP generates faster decay than the analysis model assumes, creating a systematic misspecification between the true carryover and the modeled carryover.
2. **Cumulative carryover contamination of biological response means.** In `generateData()` (line 96), carryover is added to `brmeans` cumulatively across timepoints: `brmeans[p] <- brmeans[p] +`

$\text{brmeans}[p-1] * (1/2)^{(\text{sf} * \text{tsd} / \text{t_half})}$. This recursive structure means that the biological response at off-drug timepoints depends on the entire preceding on-drug trajectory, not just the most recent on-drug value.

3. **Interaction between carryover and the correlation structure.** The original DGP generates correlated random components across timepoints, and the carryover adjustment modifies the mean structure but not the correlation matrix. This could produce model misspecification that grows with the half-life.

These candidates require empirical investigation by running the corrected 2025 simulation (with original-matching `tsd`) and comparing its power curves against the published results.

The role of `tsd_min` going forward

The `tsd_min` parameter was introduced in `pmsimstats2025` as a correction for the $t_{\text{sd}} = 0$ discontinuity. After fixing the `tsd` computation to match the original, the `tsd_min` parameter is no longer needed for that purpose.

However, `tsd_min` (or something like it) may still have value as a *design improvement* on the original. If the original's own power behavior proves to be excessive, a minimum washout interval represents a pharmacologically defensible modification: in clinical practice, there is always a nonzero gap between the last dose and the next assessment. Whether such a modification is appropriate depends on the results of the corrected simulation, and should be evaluated as a deliberate modeling choice rather than a bug fix.

Recommended Next Steps

1. **Fix `tsd` in `carryover_power_simulation.R`** to use the original interval-based accumulation logic. Verify that the corrected code produces identical `tsd` values to `buildtrialdesign()` for all four paths.
2. **Re-run the two-scenario comparison** with the corrected `tsd` (and `tsd_min = 0`) to establish a faithful reproduction of Hendrickson's simulation.
3. **Compare power curves** from the corrected 2025 simulation against the published heatmaps. If power still drops too rapidly, investigate the scale factor mismatch and cumulative carryover mechanisms listed above.
4. **Evaluate `tsd_min` as a design modification**, not a bug fix, by running a separate set of simulations with `tsd_min > 0` and comparing against the baseline.

Source Files

- **Original:** `~/prj/alz/01c-pmsimstats-orig/pmsimstats/R/buildtrialdesign.R`
- **Original vignette:** `~/prj/alz/01c-pmsimstats-orig/pmsimstats/vignettes/Produce_Publication_Results_1_generate`
- **Original analysis:** `~/prj/alz/01c-pmsimstats-orig/pmsimstats/R/lme_analysis.R`
- **2025 simulation:** `~/prj/alz/01-pmsimstats/pmsimstats2025/analysis/scripts/carryover_focus/carryover_power_sim`
- **2025 analysis document:** `~/prj/alz/01-pmsimstats/pmsimstats2025/analysis/scripts/carryover_focus/predictor_c`

Rendered on 2026-03-18 at 09:23 PDT. Source: `~/prj/alz/01-pmsimstats/pmsimstats2025/analysis/scripts/carryover_focus/t`